

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

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Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

Code:

```
File Edit Format Run Options Window Help
# Bubble sort in Python
def bubbleSort(array):

    for i in range(len(array)):

        for j in range(0, len(array) - i - 1):
            if array[j] < array[j + 1]:
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp

data = [-2, 45, 0, 11, -9]

bubbleSort(data)

print('Sorted Array in descending Order:')
print(data)
print("Kaushal Yadav")
```

Output:

```
PY
Sorted Array in descending Order:
[45, 11, 0, -2, -9]
Kaushal Yadav
```

b. Insertion sort

Code:

```
# Insertion sort in Python
def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        array[j + 1] = key

data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("yadav Kaushal")
|
```

Output:

```
PY
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
yadav Kaushal
```

c. Selection sort

Code:

```
# Selection sort in Python
def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            if array[i] > array[min_idx]:
                min_idx = i

        (array[step], array[min_idx]) = (array[min_idx], array[step])

data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in descending Order:')
print(data)
print("Kaushal==Yadav")
```

Output:

```
py
Sorted Array in descending Order:
[45, 11, 0, -2, -9]
Kaushal==Yadav
```

Curiosity Questions:

Programming Questions

1. Write a Python program to sort a list of numbers in ascending order using **Bubble Sort**.
2. Write a Python program to sort a list of numbers in ascending order using **Insertion Sort**.
3. Write a Python program to sort a list of numbers in ascending order using **Selection Sort**.

Theory Questions

4. What is sorting? Why is it required?
5. What is the basic idea of Bubble Sort?
6. How does Insertion Sort work?
7. How does Selection Sort work?
8. What is a **pass** in a sorting algorithm?
9. What is the difference between Bubble Sort and Selection Sort?
10. Which sorting algorithm is suitable when the data is already nearly sorted? Why?